THE BLOW UP FORMULA: THE RESTRICTION OF THE GYSIN MAP OF A BLOWUP IN A REGULAR CENTER IS THE TOP CHERN CLASS OF THE KERNEL OF THE CONORMAL COMPARISON

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVA (sources/stacks-chow/pinned-original-source.tex line 13808).

Allowed standard prerequisites (editorial).

- The blowup in a regular sequence is a local complete intersection morphism. More on Algebra, Lemma labelled more-algebra-lemma-blowup-regular-sequence, cited at proof lines 13827-13829. Makes the Gysin map b^! available and the statement meaningful.
- The conormal algebra of a regular immersion is the symmetric algebra. Divisors, Lemma labelled divisors-lemma-quasi-regular-immersion, cited at proof lines 13832-13837. Identifies E with the projective space bundle of the conormal sheaf, hence E -> Z smooth.
- Comparison of Gysin maps under a base change, and equality of the Gysin and the flat pullback for local complete intersection morphisms. Same chapter, Lemma labelled lemma-compare-gysin-base-change, cited at proof lines 13842-13846, and Lemma labelled lemma-lci-gysin-flat, cited at proof lines 13846-13848. Produces res(b^!) = c_top(C^vee) o pi^! and turns pi^! into pi^*.
- The exact sequences of the normal cone for a closed immersion and a smooth morphism. More on Morphisms, Lemmas labelled more-morphisms-lemma-NL-smooth, more-morphisms-lemma-get-triangle-NL and more-morphisms-lemma-NL-immersion, cited at proof lines 13852-13862. Identifies the two kernels F and C.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof.

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Source reference. Original lemma 0BIQ: sources/stacks-chow/tags.txt, source lines 14498–14498. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13829 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 063M: sources/stacks-chow/tags.txt, source lines 7836–7836. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13835 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FV9: sources/stacks-chow/pinned-original-source.tex, source lines 13738–13738. Same-chapter pointer into the author environment lemma at source line 13738; invoked at proof line 13840 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF4: sources/stacks-chow/pinned-original-source.tex, source lines 13547–13547. Same-chapter pointer into the author environment lemma at source line 13547; invoked at proof line 13846 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0D0N: sources/stacks-chow/tags.txt, source lines 16318–16318. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13853 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FV3: sources/stacks-chow/tags.txt, source lines 19798–19798. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13856 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FV2: sources/stacks-chow/tags.txt, source lines 19797–19797. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13857 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1 (Blow up formula)**.** *Let (S, δ) be as in Situation 02QL. Let $i : Z \to X$ be a regular closed immersion of schemes locally of finite type over S. Let $b : X' \to X$ be the blowing up with center Z. Picture*

$$\begin{array}{ccc} E & \xrightarrow{j} & X' \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle b} \\ Z & \xrightarrow{i} & X \end{array}$$

Assume that the Gysin map exists for b. Then we have

$$res(b^!) = c_{top}(\mathcal{F}^\vee) \circ \pi^*$$

in $A^*(E \to Z)$ where $\mathcal{F}$ is the kernel of the canonical map $\pi^*\mathcal{C}_{Z/X} \to \mathcal{C}_{E/X'}$.

Proof. Observe that the morphism b is a local complete intersection morphism by More on Algebra, Lemma 0BIQ and hence the statement makes sense. Since $Z \to X$ is a regular immersion (and hence a fortiori quasi-regular) we see that $\mathcal{C}_{Z/X}$ is finite locally free and the map $\mathrm{Sym}^*(\mathcal{C}_{Z/X}) \to \mathcal{C}_{Z/X,*}$ is an isomorphism, see Divisors, Lemma 063M. Since $E = \mathrm{Proj}(\mathcal{C}_{Z/X,*})$ we conclude that $E = \mathbf{P}(\mathcal{C}_{Z/X})$ is a projective space bundle over Z. Thus $E \to Z$ is smooth and certainly a local complete intersection morphism. Thus Lemma 0FV9 applies and we see that

$$res(b^!) = c_{top}(\mathcal{C}^\vee) \circ \pi^!$$

with $\mathcal{C}$ as in the statement there. Of course $\pi^* = \pi^!$ by Lemma 0FF4. It remains to show that $\mathcal{F}$ is equal to the kernel $\mathcal{C}$ of the map $H^{-1}(j^* NL_{X'/X}) \to H^{-1}(NL_{E/Z})$.

Since $E \to Z$ is smooth we have $H^{-1}(NL_{E/Z}) = 0$, see More on Morphisms, Lemma 0D0N. Hence it suffices to show that $\mathcal{F}$ can be identified with $H^{-1}(j^* NL_{X'/X})$. By More on Morphisms, Lemmas 0FV3 and 0FV2 we have an exact sequence

$$0 \to H^{-1}(j^* NL_{X'/X}) \to H^{-1}(NL_{E/X}) \to \mathcal{C}_{E/X'} \to \ldots$$

By the same lemmas applied to $E \to Z \to X$ we obtain an isomorphism $\pi^*\mathcal{C}_{Z/X} = H^{-1}(\pi^* NL_{Z/X}) \to H^{-1}(NL_{E/X})$. Thus we conclude. $\square$